

ALFAAZ37079

## 7-Oxabicyclo[4.1.0]hept-3-ylmethyl-7-oxabicyclo[4.1.0]heptane-3-carboxylate

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 7-Oxabicyclo[4.1.0]hept-3-ylmethyl-7-oxabicyclo[4.1.0]heptane-3-carboxylate  
7-Oxabicyclo[4.1.0]hept-3-ylmethyl-7-oxabicyclo[4.1.0]heptane-3-carboxylate

**Cat No. :** Z37079  
**CAS No** 2386-87-0

**Supplier** Avocado Research Chemicals Ltd.  
(Part of Thermo Fisher Scientific)  
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Lancashire, LA3 2XY,  
United Kingdom  
Office Tel: +44 (0) 1524 850506  
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Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Solid

**Appearance**  
No information available

**Odor**  
No information available

**Emergency Overview**  
May be harmful if swallowed.

#### Classification of the substance or mixture

Acute Oral Toxicity

Category 5

#### Label Elements

None required

#### **Hazard Statements**

H303 - May be harmful if swallowed

#### **Precautionary Statements**

##### **Prevention**

P270 - Do not eat, drink or smoke when using this product

##### **Response**

P301 + P312 - IF SWALLOWED: Call a POISON CENTER or doctor if you feel unwell

##### **Storage**

P403 - Store in a well-ventilated place

## 7-Oxabicyclo[4.1.0]hept-3-ylmethyl-7-oxabicyclo[4.1.0]heptane-3-carboxylate

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

None identified.

**Health Hazards**

May be harmful if swallowed.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants.

This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component                                                              | CAS No    | Weight % |
|------------------------------------------------------------------------|-----------|----------|
| 3,4-Epooxycyclohexanecarboxylic acid (3,4-epoxycyclohexylmethyl) ester | 2386-87-0 | <=100    |

**SECTION 4. FIRST AID MEASURES****Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

**Inhalation**

Remove to fresh air. Get medical attention immediately if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Most important symptoms and effects**

None reasonably foreseeable.

**Self-Protection of the First Aider**

No special precautions required.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES**

## 7-Oxabicyclo[4.1.0]hept-3-ylmethyl-7-oxabicyclo[4.1.0]heptane-3-carboxylate

**Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation. Avoid dust formation.

**Storage**

Keep container tightly closed in a dry and well-ventilated place.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

**Exposure Controls****Engineering Measures**

None under normal use conditions.

**Personal protective equipment****Eye Protection**

Wear safety glasses with side shields (or goggles) (European standard - EN 166)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Neoprene       | recommendations   |                 |             |                       |
| Natural rubber |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Long sleeved clothing

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|                                        |                                                                                                                                                                                                        |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Respiratory Protection</b>          | No protective equipment is needed under normal use conditions.                                                                                                                                         |
| <b>Large scale/emergency use</b>       | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced<br><b>Recommended Filter type:</b> Particle filter |
| <b>Small scale/Laboratory use</b>      | Maintain adequate ventilation                                                                                                                                                                          |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                 |
| <b>Environmental exposure controls</b> | No information available.                                                                                                                                                                              |

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>                              |                          |                                          |
| <b>Physical State</b>                          | Solid                    |                                          |
| <b>Odor</b>                                    | No information available |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>pH</b>                                      | Not applicable           |                                          |
| <b>Melting Point/Range</b>                     | No data available        |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | No information available |                                          |
| <b>Flash Point</b>                             | No information available | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | Not applicable           | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available |                                          |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Vapor Pressure</b>                          | No data available        |                                          |
| <b>Vapor Density</b>                           | Not applicable           | Solid                                    |
| <b>Specific Gravity / Density</b>              | No data available        |                                          |
| <b>Bulk Density</b>                            | No data available        |                                          |
| <b>Water Solubility</b>                        | No information available |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Component</b>                               | <b>log Pow</b>           |                                          |
| 3,4-Epoxy cyclohexanecarboxylic acid           | 1.34                     |                                          |
| (3,4-epoxycyclohexylmethyl) ester              |                          |                                          |
| <b>Autoignition Temperature</b>                | No data available        |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>Viscosity</b>                               | Not applicable           | Solid                                    |
| <b>Explosive Properties</b>                    | No information available |                                          |
| <b>Oxidizing Properties</b>                    | No information available |                                          |
| <b>Molecular Weight</b>                        | 252.34                   |                                          |

## SECTION 10. STABILITY AND REACTIVITY

|                                 |                                 |
|---------------------------------|---------------------------------|
| <b>Stability</b>                | Stable under normal conditions. |
| <b>Hazardous Reactions</b>      | None under normal processing.   |
| <b>Hazardous Polymerization</b> | No information available.       |
| <b>Conditions to Avoid</b>      | None known.                     |

## 7-Oxabicyclo[4.1.0]hept-3-ylmethyl-7-oxabicyclo[4.1.0]heptane-3-carboxylate

**Materials to avoid** Strong oxidizing agents.

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

**SECTION 11. TOXICOLOGICAL INFORMATION****Product Information****(a) acute toxicity;**

| Component                                                                 | LD50 Oral                 | LD50 Dermal                   | LC50 Inhalation |
|---------------------------------------------------------------------------|---------------------------|-------------------------------|-----------------|
| 3,4-Epoxy cyclohexanecarboxylic acid<br>(3,4-epoxycyclohexylmethyl) ester | LD50 = 5000 mg/kg ( Rat ) | LD50 = 23600 mg/kg ( Rabbit ) |                 |

**(b) skin corrosion/irritation;** No data available

**(c) serious eye damage/irritation;** No data available

**(d) respiratory or skin sensitization;**

**Respiratory** No data available  
**Skin** No data available

**(e) germ cell mutagenicity;** No data available

**(f) carcinogenicity;** No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

**(g) reproductive toxicity;** No data available

**(h) STOT-single exposure;** No data available

**(i) STOT-repeated exposure;** No data available

**Target Organs** No information available.

**(j) aspiration hazard;** Not applicable  
Solid

**Symptoms / effects, both acute and delayed** No information available

**SECTION 12. ECOLOGICAL INFORMATION****Ecotoxicity effects**

| Component                                                                 | Freshwater Fish                                               | Water Flea | Freshwater Algae | Microtox |
|---------------------------------------------------------------------------|---------------------------------------------------------------|------------|------------------|----------|
| 3,4-Epoxy cyclohexanecarboxylic acid<br>(3,4-epoxycyclohexylmethyl) ester | LC50: = 24 mg/L, 96h<br>flow-through<br>(Oncorhynchus mykiss) |            |                  |          |

**Persistence and Degradability** No information available

## 7-Oxabicyclo[4.1.0]hept-3-ylmethyl-7-oxabicyclo[4.1.0]heptane-3-carboxylate

**Bioaccumulative Potential** No information available

| Component                                                                | log Pow | Bioconcentration factor (BCF) |
|--------------------------------------------------------------------------|---------|-------------------------------|
| 3,4-Epoxycyclohexanecarboxylic acid<br>(3,4-epoxycyclohexylmethyl) ester | 1.34    | No data available             |

**Mobility in soil** No information available**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors**Persistent Organic Pollutant** This product does not contain any known or suspected substance**Ozone Depletion Potential** This product does not contain any known or suspected substance**SECTION 13. DISPOSAL CONSIDERATIONS****Waste from Residues/Unused Products** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.**Contaminated Packaging** Empty remaining contents. Dispose of in accordance with local regulations. Do not re-use empty containers.**Other Information** Waste codes should be assigned by the user based on the application for which the product was used.**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport** Not Regulated**IMDG/IMO** Not regulated**IATA** Not regulated**Special Precautions for User** No special precautions required**SECTION 15. REGULATORY INFORMATION****International Inventories**

China, X = listed, Australia, U.S.A. (TSCA), Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS).

| Component                                                                 | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|---------------------------------------------------------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| 3,4-Epoxycyclohexane carboxylic acid<br>(3,4-epoxycyclohexylmethyl) ester | -                                                   | -                                       | X    | X     | 219-207-4 | X    | X   | X     | X    | X    | X    | KE-27441 |

**National Regulations**

## 7-Oxabicyclo[4.1.0]hept-3-ylmethyl-7-oxabicyclo[4.1.0]heptane-3-carboxylate

## SECTION 16. OTHER INFORMATION

**Prepared By** Health, Safety and Environmental Department  
**Revision Date** 12-May-2024  
**Revision Summary** New emergency telephone response service provider.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

**Legend**

|                                                                                                                              |                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>CAS</b> - Chemical Abstracts Service                                                                                      | <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory                   |
| <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances | <b>DSL/NDL</b> - Canadian Domestic Substances List/Non-Domestic Substances List                   |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                                                    | <b>ENCS</b> - Japanese Existing and New Chemical Substances                                       |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                                             | <b>AICS</b> - Australian Inventory of Chemical Substances                                         |
| <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              | <b>NZIoC</b> - New Zealand Inventory of Chemicals                                                 |
| <b>WEL</b> - Workplace Exposure Limit                                                                                        | <b>TWA</b> - Time Weighted Average                                                                |
| <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists                                                     | <b>IARC</b> - International Agency for Research on Cancer                                         |
| <b>DNEL</b> - Derived No Effect Level                                                                                        | <b>PNEC</b> - Predicted No Effect Concentration                                                   |
| <b>RPE</b> - Respiratory Protective Equipment                                                                                | <b>LD50</b> - Lethal Dose 50%                                                                     |
| <b>LC50</b> - Lethal Concentration 50%                                                                                       | <b>EC50</b> - Effective Concentration 50%                                                         |
| <b>NOEC</b> - No Observed Effect Concentration                                                                               | <b>POW</b> - Partition coefficient Octanol:Water                                                  |
| <b>PBT</b> - Persistent, Bioaccumulative, Toxic                                                                              | <b>vPvB</b> - very Persistent, very Bioaccumulative                                               |
| <b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association                         | <b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code |
| <b>ADR</b> - European Agreement Concerning the International Carriage of Dangerous Goods by Road                             | <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships               |
| <b>OECD</b> - Organisation for Economic Co-operation and Development                                                         | <b>ATE</b> - Acute Toxicity Estimate                                                              |
| <b>BCF</b> - Bioconcentration factor                                                                                         | <b>VOC</b> - (Volatile Organic Compound)                                                          |

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>  
 Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

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**End of Safety Data Sheet**